REB, The Book
Example 6.6.6 Solution

Example 6.6.4 shows conditions where three steady states are possible when reaction (1) takes place in an adiabatic CSTR. Specifically, the reactor volume is 500 cm^3 and the feed is 1.0 cm^{-3} s^{-1} of a solution containing 0.015 mol cm^{-3} each of A and B at 50 °C. The heat capacity is 0.35 cal g^{-1} K^{-1}, and the density is 0.93 gm cm^{-3}. The heat of reaction is –20 kJ mol^{-1}, and the rate is given by equation (2) with the pre-exponential factor equal to 3.24 x 10^{12} cm^3 mol^{-1} s^{-1} and the activation energy equal to 105 kJ mol^{-1}. The steady-state conversions were 0.03, 39.9, and 97.5%, and the corresponding outlet temperatures were 50, 183, and 265 °C.

$$A + B \rightarrow Y + Z \tag{1}$$

$$r_1 = k_1 C_A C_B \tag{2}$$

The outlet temperature for this reactor was plotted as a function of the feed temperature, and that graph is reproduced here as Figure 1, with three steady states labeled "S_{high}," "U_{mid}," and "S_{low}" Steady states S_{high} (97.5% conversion) and S_{low} (0.03% conversion) are stable while steady state U_{mid} (39.3% conversion) is unstable.

To show that steady state U_{mid} is unstable, plot the outlet temperature *vs.* time if the temperature of a CSTR at steady state U_{mid} is perturbed +1 °C, and by -1 °C holding all of the inputs constant. To show the two other steady states are stable, plot the outlet temperature *vs*. time if the temperature of a CSTR at each of those steady states is perturbed +20 °C, and by -20 °C holding all of the inputs constant.

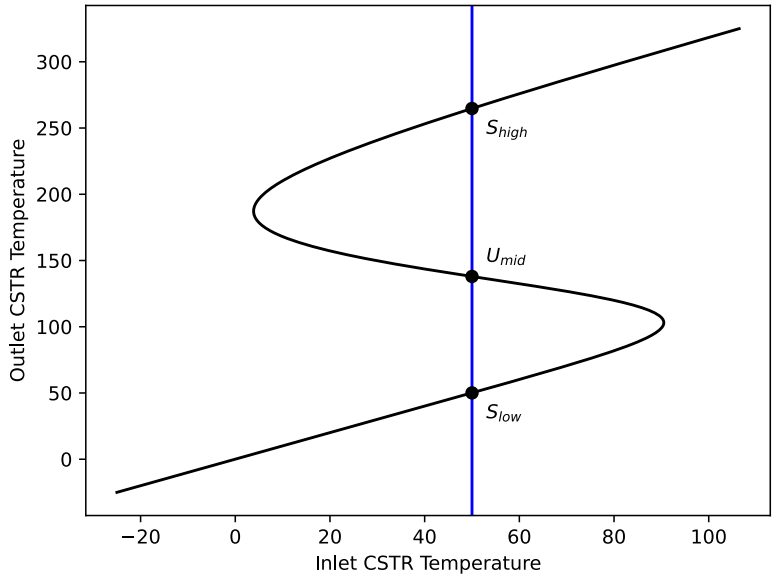


Figure 1. Steady-state temperatures prior to perturbation.

Assignment Summary

Reaction

$$A + B \rightarrow Y + Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A C_B \tag{2}$$

Reactor: Transient, adiabatic CSTR

Given Constants: V = 500 cm^3, $\dot{V}_{in}$ = 1.0 cm^3 s^{-1}, $C_{A,in}$ = 0.015 mol cm^{-3}, $C_{B,in}$ = 0.015 mol cm^{-3}, T_{in} = (50 + 273.15) K, $\tilde{C}_p$ = 0.35 cal g^{-1} K^{-1}, ρ = 0.93 gm cm^{-3}, ΔH_1 = –20 kJ mol^{-1}, $k_{0,1}$ = 3.24 x 10^{12} cm^3 mol^{-1} s^{-1}, and E_1 = 105.0 kJ mol^{-1}

Parameters: $\underline{f}_{A,0} = [0.03, 39.9, 97.5]$%, $\underline{T} = ([50, 183, 265] + 273.15)$ K, and $\underline{\Delta T} = [20, 1, 20]$ K

Deliverables: Plots of $\underline{T}$ vs. $\underline{t}$ following ± 1 °C perturbations from the U_{mid} steady state and following ± 20 °C perturbations form the S_{high} and S_{low} steady states.

Completion Strategy

The outlet temperature *versus* time can be calculated by solving the CSTR design equations. Here, the design equations need to be solved two times for each steady state. One solution is for a positive perturbation, and one is for a negative perturbation.

Formulation of the Calculations

CSTR Model Function

CSTR Design Equations:

$$\frac{d\dot{n}_A}{dt} = \frac{\dot{V}_{in}}{V}\left(\dot{n}_{A,in} - \dot{n}_A - Vr_1\right) \tag{3}$$

$$\frac{d\dot{n}_B}{dt} = \frac{\dot{V}_{in}}{V}\left(\dot{n}_{B,in} - \dot{n}_B - Vr_1\right) \tag{4}$$

$$\frac{d\dot{n}_Y}{dt} = \frac{\dot{V}_{in}}{V}\left(-\dot{n}_Y + Vr_1\right) \tag{5}$$

$$\frac{d\dot{n}_Z}{dt} = \frac{\dot{V}_{in}}{V}\left(-\dot{n}_Z + Vr_1\right) \tag{6}$$

$$\frac{dT}{dt} = \frac{-\tilde{C}_p\rho\dot{V}_{in}\left(T - T_{in}\right) - Vr_1\Delta H_1}{V\tilde{C}_p\rho} \tag{7}$$

CSTR Model Variables: $\underline{t}$, $\underline{\dot{n}}_A$, $\underline{\dot{n}}_B$, $\underline{\dot{n}}_Y$, $\underline{\dot{n}}_Z$, and $\underline{T}$

Additional Computable Unknowns: $\dot{n}_{A,in}$, r_1, k_1, C_A, $\dot{V}$, C_B, and $\dot{n}_{B,in}$

$$\dot{n}_{i,in} = \dot{V}_{in} C_{i,in} \qquad i = A \text{ and } B \tag{8}$$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{9}$$

$$C_i = \frac{\dot{n}_i}{\dot{V}} \qquad i = A \text{ and } B \tag{10}$$

$$\dot{V} = \dot{V}_{in} \tag{11}$$

Additional Non-Computable Unknowns: $f_{A,0}$, T_0, and ΔT

IVODE Solver Requirements:

1. Initial values of the CSTR model variables (provided as an argument to the CSTR model function)
2. Stopping criterion (provided as an argument to the CSTR model function)
3. Derivatives function that
   a. receives values of the reactor model variables at the start of an integration step
   b. calculates the additional unknowns using equations (2) and (8) - (11)
   c. evaluates and returns the design equation derivatives using equations (3) - (7)

Deliverables Function

1. Loop through the three sets of parameter values
   a. for each set of parameters, n
      i. define the initial values for a positive perturbation of the temperature

$$t\Big|_{t=0} = 0 \tag{12}$$

$$\dot{n}_A\Big|_{t=0} = \dot{n}_{A,in}\left(1 - f_{A,0,n}\right) \tag{13}$$

$$\dot{n}_B\Big|_{t=0} = \dot{n}_{B,in} - f_{A,0,n}\dot{n}_{A,in} \tag{14}$$

$$\dot{n}_i\Big|_{t=0} = f_{A,0,n}\dot{n}_{A,in} \qquad i = Y \text{ and } Z \tag{15}$$

$$T\Big|_{t=0} = T_n + \Delta T_n \tag{16}$$

ii. define the stopping criterion

$$t = 1 \text{ h} \tag{17}$$

iii. call the CSTR model function, passing the initial values and stopping criterion as arguments to get $\underline{t}$ and $\underline{T}$

iv. repeat using a negative perturbation of the temperature for the initial value of the temperature

$$T\Big|_{t=0} = T_n - \Delta T_n \tag{18}$$

Implementation of the Calculations

The Python script, example_6_6_6.py, and Matlab function, example_6_6_6.m, that were written to perform the calculations accompany this solution.

Results and Discussion

The calculations were performed as described above. The value of t_f was adjusted so that the transients filled the graphs. Figure 1 shows the temperature as a function of time following perturbation from the unstable steady state by plus and minus 1 °C holding all of the inputs constant. Since none of the reactor inputs changed, the temperature would have returned to 138 °C had that been a stable steady state. Instead, the figure shows that following a 1 °C increase in the reacting fluid temperature, it continues to increase and stabilizes at the high temperature steady

state. Following a 1°C decrease in the reacting fluid temperature, it continues to decrease and stabilizes at the low temperature steady state.

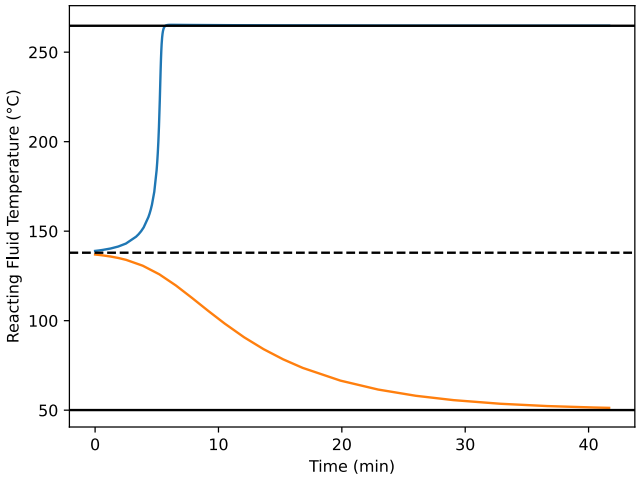


Figure 1. Reacting fluid temperature following a +1 °C (blue) and a -1 °C (orange) perturbation from the unsteady state. The dashed black line represents the unstable steady state, and the solid black lines represent the stable steady states.

The duration of the transients in Figure 1 are different. This is likely because following the positive temperature perturbation the temperature is rising causing the reaction rate to rise also. Due to the faster reaction rates, the changes occur rapidly. Following the negative temperature perturbation, the temperature is decreasing and changes occur more slowly.

For both the high temperature, Figure 2, and low temperature, Figure 3, steady states, following much larger perturbations of ± 20 °C the system returns to its original steady state. At the high temperature, the response briefly moves farther from the steady-state before reaching a maximum deviation and then returning to the steady state.

The brief increase in the deviation following perturbation from the high temperature steady state is likely due to concentration effects since only the temperature is perturbed. That causes an immediate jump in the rate, after which the

changing composition and temperature have opposing effects. For a brief period, the effect of increasing temperature predominates, but after a short interval, the two effects become equal and the temperature deviation reaches a maximum. From that point on, the concentration effects predominate and the deviation steadily decreases. Apparently concentration effects predominate immediately after perturbation from the low temperature steady state.

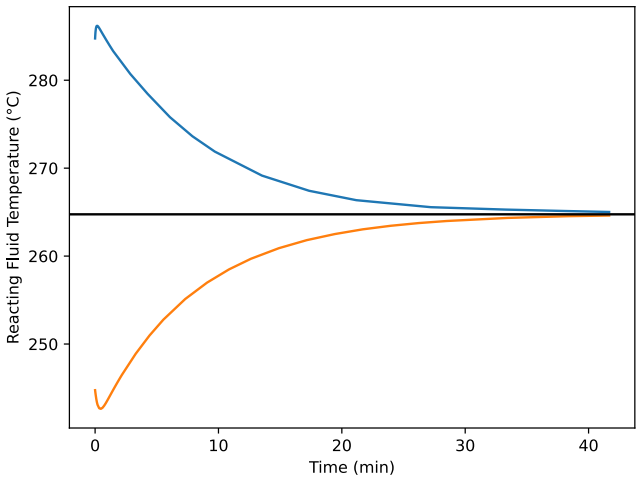


Figure 2. Reacting fluid temperature following a +1 °C (blue) and a -1 °C (orange) perturbation from the high temperature (black) steady state.

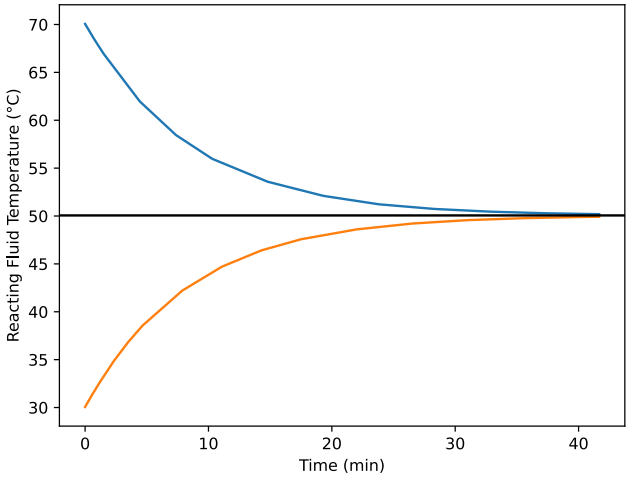


Figure 3. Reacting fluid temperature following a +1 °C (blue) and a -1 °C (orange) perturbation from the low temperature (black) steady state.

Summary

A 1°C change of the reacting fluid temperature from the unstable steady state initiates transient operation leading to one of the stable steady states. This shows confirms that the initial steady state is, indeed, unstable. In contrast, a 20 °C change of the reacting fluid temperature from either of the stable steady states initiates transient operation leading back to the original steady state, confirming that these initial steady states are stable.